

LITERATURE REVIEW REPORT

AI-BASED FAKE PROFILE DETECTION

1. INTRODUCTION

Fake-account detection has developed from simple rule-based moderation toward statistical learning, behavioral analysis, graph mining and deep learning.

2. PROFILE-BASED APPROACHES

Profile-based systems analyze account age, follower/following relationships, profile completeness, username properties, biography fields, verification status and posting volume. Their advantage is relatively low computational complexity. Their weakness is that sophisticated fake profiles can imitate normal profile characteristics.

3. BEHAVIOR-BASED APPROACHES

Behavioral systems analyze activity frequency, time intervals, burst patterns, interaction ratios and changes over time. These approaches can identify accounts whose operational behavior differs from normal users.

4. GRAPH-BASED APPROACHES

Social networks naturally form graphs. Users are nodes and interactions are edges. Graph methods can detect suspicious clusters, coordinated activity and abnormal connection patterns. Graph Neural Networks provide an advanced direction for future work.

5. CONTENT-BASED APPROACHES

Natural-language processing can examine posts, biographies and messages. Transformer models may identify spam-like or coordinated content patterns. Content analysis introduces additional privacy and language-related challenges.

6. MACHINE LEARNING

Logistic Regression offers interpretability. Decision Trees provide readable rules. Random Forest and Gradient Boosting are strong tabular-data approaches. Neural networks can model complex relationships but may require more data and computational resources.

7. RESEARCH GAP

A practical system should combine multiple signals, provide explanations, expose predictions through an API, and support human review. The proposed project integrates these requirements into a complete lifecycle.

8. INNOVATION

The main innovation is the combination of behavioral intelligence, ensemble ML, risk scoring and explainability within a deployable architecture.

9. CONCLUSION

The literature indicates that no single feature is sufficient for reliable fake-profile detection. Combining complementary signals and continuously evaluating the model is therefore important.